

Name :	Siddhanth Mungekar
Roll No	23102B0030
Branch	CMPN/B
NPTEL Week	7

Screenshots	<pre> R - R 4.6.1 - ~/Documents/R Programming/Assignment_4/ > # Store a sequence > x = seq(1, 50, 1/2) > x [1] 1.0 1.5 2.0 2.5 3.0 3.5 4.0 4.5 5.0 5.5 6.0 6.5 7.0 7.5 8.0 8.5 9.0 [18] 9.5 10.0 10.5 11.0 11.5 12.0 12.5 13.0 13.5 14.0 14.5 15.0 15.5 16.0 16.5 17.0 17.5 [35] 18.0 18.5 19.0 19.5 20.0 20.5 21.0 21.5 22.0 22.5 23.0 23.5 24.0 24.5 25.0 25.5 26.0 [52] 26.5 27.0 27.5 28.0 28.5 29.0 29.5 30.0 30.5 31.0 31.5 32.0 32.5 33.0 33.5 34.0 34.5 [69] 35.0 35.5 36.0 36.5 37.0 37.5 38.0 38.5 39.0 39.5 40.0 40.5 41.0 41.5 42.0 42.5 43.0 [86] 43.5 44.0 44.5 45.0 45.5 46.0 46.5 47.0 47.5 48.0 48.5 49.0 49.5 50.0 > > y = 2*x > y [1] 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 [23] 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 [45] 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 [67] 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 [89] 90 91 92 93 94 95 96 97 98 99 100 > > # Index vector > x = c(9,8,7,6) > ind = seq(along=x) > ind [1] 1 2 3 4 > > x[ind[2]] [1] 8 > > # Current date and time > Sys.time() [1] "2026-09-03 15:10:11 IST" > Sys.Date() [1] "2026-09-03" > > # --- Sequences of dates --- > > seq(+ as.Date("2010-01-01"), + as.Date("2017-01-01"), + by = "years" +) [1] "2010-01-01" "2011-01-01" "2012-01-01" "2013-01-01" "2014-01-01" "2015-01-01" [7] "2016-01-01" "2017-01-01" > </pre>
	<pre> R - R 4.6.1 - ~/Documents/R Programming/Assignment_4/ > seq(+ as.Date("2017-01-01"), + by = "days", + length = 6 +) [1] "2017-01-01" "2017-01-02" "2017-01-03" "2017-01-04" "2017-01-05" "2017-01-06" > > seq(+ as.Date("2017-01-01"), + by = "months", + length = 6 +) [1] "2017-01-01" "2017-02-01" "2017-03-01" "2017-04-01" "2017-05-01" "2017-06-01" > > seq(+ as.Date("2017-01-01"), + by = "years", + length = 6 +) [1] "2017-01-01" "2018-01-01" "2019-01-01" "2020-01-01" "2021-01-01" "2022-01-01" > > startdate = as.Date("2016-1-1") > enddate = as.Date("2017-1-1") > > out = seq(enddate, startdate, by = "-1 month") > out [1] "2017-01-01" "2016-12-01" "2016-11-01" "2016-10-01" "2016-09-01" "2016-08-01" [7] "2016-07-01" "2016-06-01" "2016-05-01" "2016-04-01" "2016-03-01" "2016-02-01" [13] "2016-01-01" > > # --- Alphabet sequences --- > > letters [1] "a" "b" "c" "d" "e" "f" "g" "h" "i" "j" "k" "l" "m" "n" "o" "p" "q" "r" "s" "t" "u" "v" [23] "w" "x" "y" "z" > letters[1:3] [1] "a" "b" "c" > letters[3:1] [1] "c" "b" "a" > letters[21:23] [1] "u" "v" "w" > letters[2] [1] "b" > </pre>

```

RStudio - R 4.6.1 - ~/Documents/R Programming/Assignment_4/
Console Terminal Background Jobs
> LETTERS
[1] "A" "B" "C" "D" "E" "F" "G" "H" "I" "J" "K" "L" "M" "N" "O" "P" "Q" "R" "S" "T" "U" "V"
[23] "W" "X" "Y" "Z"
> LETTERS[1:3]
[1] "A" "B" "C"
> LETTERS[3:1]
[1] "C" "B" "A"
> LETTERS[21:23]
[1] "U" "V" "W"
> LETTERS[2]
[1] "B"
>
>
> # --- rep() ---
>
> help("rep")
>
> rep(3.5, times=10)
[1] 3.5 3.5 3.5 3.5 3.5 3.5 3.5 3.5 3.5 3.5
> rep(1:4, 2)
[1] 1 2 3 4 1 2 3 4
>
> x = 1:4
> x
[1] 1 2 3 4
> rep(x, times=3)
[1] 1 2 3 4 1 2 3 4 1 2 3 4
> rep(x, each=3)
[1] 1 1 1 2 2 2 3 3 3 4 4 4
>
> rep(1:4, each=2)
[1] 1 1 2 2 3 3 4 4
> rep(1:4, each=2, times=3)
[1] 1 1 2 2 3 3 4 4 1 1 2 2 3 3 4 4 1 1 2 2 3 3 4 4
> rep(1:4, times=3, each=2)
[1] 1 1 2 2 3 3 4 4 1 1 2 2 3 3 4 4 1 1 2 2 3 3 4 4
>
> # Different number of repetitions
> rep(1:4, 2:5)
[1] 1 1 2 2 2 3 3 3 3 4 4 4 4 4
>
> ans = seq(from=2, to=8, by=2)
> ans
[1] 2 4 6 8
> rep(1:4, ans)
[1] 1 1 2 2 2 2 3 3 3 3 3 3 3 3 4 4 4 4 4 4 4 4 4 4

```

```

> # Repeating a matrix
> x = matrix(nrow=2, ncol=2, data=1:4, byrow=T)
> x
      [,1] [,2]
[1,]    1    2
[2,]    3    4
> rep(x, 3)
[1] 1 3 2 4 1 3 2 4 1 3 2 4
>
> # Repeating characters
> rep(c("a","b","c"), 2)
[1] "a" "b" "c" "a" "b" "c"
> rep(c("apple","banana","cake"), 2)
[1] "apple" "banana" "cake" "apple" "banana" "cake"
>
> # Pre-specified length
> rep(2, length.out=5)
[1] 2 2 2 2 2
> rep(2, length=5)
[1] 2 2 2 2 2
> rep(c(2,3), length=5)
[1] 2 3 2 3 2
> rep(c(2,3,4), length=5)
[1] 2 3 4 2 3
>
> rep("apple", length=5)
[1] "apple" "apple" "apple" "apple" "apple"
> rep(c("a","b","c"), length=2)
[1] "a" "b"
> rep(c("a","b","c"), length=5)
[1] "a" "b" "c" "a" "b"
>
>
> # --- Sorting ---
>
> y = c(8,5,7,6)
> y
[1] 8 5 7 6
> sort(y)
[1] 5 6 7 8
> sort(y, decreasing=TRUE)
[1] 8 7 6 5

```

ConsoleTerminalBackground Jobs

R 4.6.1 · ~/Documents/R Programming/Assignment_4/

```
> # --- Ordering ---
>
> y = c(9,8,5,7,6)
> y
[1] 9 8 5 7 6
> order(y)
[1] 3 5 4 2 1
> order(y, decreasing=TRUE)
[1] 1 2 4 5 3
>
>
> # --- Mode ---
>
> mode(2.432)
[1] "numeric"
> mode(c(3,4,5,6,7,8))
[1] "numeric"
> mode("India")
[1] "character"
> mode(c("India","CANADA"))
[1] "character"
>
> mode(factor(c("UP","MP"))))
[1] "numeric"
> mode(list("India","USA"))
[1] "list"
> mode(data.frame(x=1:2, y=c("India","USA")))
[1] "list"
> mode(print)
[1] "function"
>
>
> # --- Lists ---
>
> x1 = matrix(nrow=2, ncol=2, data=1:4, byrow=T)
> x2 = matrix(nrow=2, ncol=2, data=5:8, byrow=T)
>
> x1
      [,1] [,2]
[1,]    1    2
[2,]    3    4
> x2
      [,1] [,2]
[1,]    5    6
[2,]    7    8
> x1 + x2
      [,1] [,2]
[1,]    6    8
[2,]   10   12
>
> # Changing a matrix element to character
> x1[2,1] = "hello"
> x1
      [,1] [,2]
[1,]    "1"    "2"
[2,] "hello"    "4"
> x1 + x2
Error in x1 + x2 : non-numeric argument to binary operator
Show Traceback
Rerun with Debug
> |
```

Data		
d	num [1:3, 1:3] 5 0 0 0 5 0 0 0 5	
ⓘ warpbreaks	54 obs. of 3 variables	
x	int [1:2, 1:2] 1 3 2 4	
x1	chr [1:2, 1:2] "1" "hello" "2" "4"	
x2	int [1:2, 1:2] 5 7 6 8	
xt	int [1:2, 1:4] 1 2 3 4 5 6 7 8	
Values		
ans	num [1:4] 2 4 6 8	
child	chr [1:3] "child1" "child2" "child3"	
drink	chr [1:4] "coffee" "lemonade" "tea" "juice"	
enddate	2017-01-01 UTC	
i	11	
ind	int [1:4] 1 2 3 4	
out	Date[1:13], format: "2017-01-01" "2016-12-01" "2016-11-01" "2016-10-...	
startdate	2016-01-01 UTC	
sweet	chr [1:3] "sweet1" "sweet2" "sweet3"	
y	num [1:5] 9 8 5 7 6	
Functions		
excount	function (x)	
sumfunction	function ()	

